

Lesson plan

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Network Security	
Course code	CS3403	
Semester:	VI	
Names of the Course Instructor(s)	Prof. Cynthia, Prof. Evlin Vidyulatha, Prof. Ishita Chakraborty, Prof. Lokesh J.K., Prof. Neha Najeem, Prof. Sheba Pari, Prof. Sonam V Maju, Prof. Sumathi H.R., Prof. Sunil Kumar J.	
Course credit	No. of hours per week (3+0+2)	Total no. of Teaching hours
04	05 Hours	75 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	1	Introduction to Network Security and OSI Model	Interactive PPT	Stallings Ch.1
2.		TCP Connection Management and ISN	TCP Three-Way Handshake Simulation using Wireshark	Peterson Ch.6
3.		TCP State Transitions and Options	Discussion on how attackers exploit state transitions (e.g., SYN flood).	Peterson Ch.6
4.		LAB 1- Introduction: Recap of Cisco Packet Tracer and network building using a simple network	Hands-on using Cisco Packet Tracer	
5.				
6.		TCP Flow Control and Window Management	Students calculate throughput under different window sizes.	Stallings (Network Security) Ch.9
7.		Congestion Control Protocols (DCCP, QUIC)	Compare TCP vs DCCP vs QUIC using a comparison table:	Peterson Ch.6
8.		DASH	Case study & comparison	Peterson Ch.7
9.		LAB 2: Working principles and operation of NAT using CISCO packet tracer	Hands-on using Cisco Packet Tracer	
10.				

11.		DASH, and its relevance with respect to applications.	Case study & comparison	Peterson Ch.7
12.	2	Introduction to Routing Protocols	PPT-based explanation with real-world network examples.	Peterson Ch.4
13.		Distance Vector Routing (RIP)	Chalk-and-talk with routing table simulation activity.	Peterson Ch.4
14.		LAB 3: Server and Client Interactions Using TCP Sockets	Hands-on programming using WSL, Windows	
15.				
16.		Open Shortest Path First (OSPF) – Part 1	PPT with step-by-step illustration of Dijkstra's algorithm.	Peterson Ch.4
17.		Open Shortest Path First (OSPF) – Part 2	Case-study-based learning using multi-area OSPF diagrams.	Peterson Ch.4
18.		Router Implementation and Performance	PPT followed by discussion on router performance metrics.	Peterson Ch.4
19.		LAB 4: Encrypted data over TCP Socket using OpenSSL	Hands-on programming using WSL, Windows	
20.				
21.		Interdomain Routing and BGP – Part 1	Interactive PPT	Peterson Ch.4
22.		Border Gateway Protocol (BGP) – Part 2	Case study discussion on BGP routing and security issues.	Peterson Ch.4
23.		IPv6 Introduction and Address Space	PPT with IPv4–IPv6 comparison examples.	Peterson Ch.4
24.		Hardware Lab Introduction:	Hands-on using ICX7150 switches/routers	
25.		Introduction to Lab: Switch mode and Router mode of ICX7150 devices		
26.		IP Multicast and Routing	PPT and diagram-based explanation of multicast communication.	Peterson Ch.7
27.		Software Defined Networking (SDN) Concepts	PPT with traditional vs SDN architecture comparison.	Peterson Ch.8

28.		SDN Controller and Data Plane	Demonstration-based learning using SDN flow diagrams.	Peterson Ch.8
29.		Lab 5: Introduction to hardware lab by using Telnet and simple ping between the same networks.	Hands-on using ICX7150 switches/routers	
30.				
31.		Practice Session – Routing Protocols	Pen & Paper	
32.		VPN and Tunneling Mechanisms	PPT-based explanation with secure communication diagrams.	Stallings (Network Security) Ch.10
33.		Mobile IP and Routing	PPT with step-by-step mobility and handoff illustrations.	Peterson Ch.9
34.		Lab 6: Construct a VLAN network where communication between same VLAN should be possible and different VLANs should not be possible.	Hands-on using ICX7150 switches/routers	
35.				
36.		Remote Procedure Call (RPC)	Chalk-and-talk supported by client-server workflow diagrams.	Peterson Ch.5
37.	3	QoS Parameters and Management	PPT with real-time traffic prioritization examples.	Peterson Ch.5
38.		RED and Queue Management	PPT with congestion control and queue behavior visualization.	Peterson Ch.6
39.		Lab 7: Construct an Inter-VLAN network and demonstrate communication between different VLANs.	Hands-on using ICX7150 switches/routers	
40.				
41.		DiffServ Architecture and Implementation	Diagram-based PPT explaining traffic classification and marking.	Peterson Ch.6
42.		VPN Protocols and Security	PPT with comparative analysis of IPSec, SSL, and L2TP.	Stallings (Network Security) Ch.10

43.		Practice Session – VPN and QoS	Hands-on lab using Packet Tracer for VPN and QoS configuration.	Combined
44.	4	Lab 8: Demonstrate Access Control List in hardware lab.	Hands-on using ICX7150 switches/routers	
45.				
46.		Introduction to Cryptography and CIA Triad	PPT-based explanation with real-world security breach examples.	Stallings (Cryptography) Ch.1
47.		Symmetric Encryption (DES)	Chalk-and-talk with step-by-step block cipher illustration.	Stallings (Cryptography) Ch.3
48.		Advanced Encryption Standard (AES)	PPT with round-based encryption process diagrams.	Stallings (Cryptography) Ch.5
49.		Lab 9: Configure Router on a Stick configuration using the Hardware lab.	Hands-on using ICX7150 switches/routers	
50.				
51.		Public-Key Cryptography and RSA	PPT explaining key generation and encryption–decryption flow.	Stallings (Cryptography) Ch.9
52.		Cryptographic Hash Functions	PPT with data integrity and authentication use cases.	Stallings (Cryptography) Ch.11
53.		SHA-512 and Secure Hashing	PPT with message digest computation example.	Stallings (Cryptography) Ch.11
54.		Lab 10: Demonstrate the working of OSPF	Hands-on using ICX7150 switches/routers	
55.				
56.		Cipher Block Modes of Operation	PPT with ECB, CBC, CFB, OFB, and CTR mode comparisons.	Stallings (Cryptography) Ch.6
57.		Introduction to Post-Quantum Cryptography	PPT followed by discussion on quantum computing threats to RSA and ECC.	Stallings (Cryptography) Ch.17
58.	5	Network Attacks Classification and Basics	Case-study–based learning with attack scenarios	Stallings (Network Security) Ch.1
59.		Lab 11: DHCP snooping	Hands-on using ICX7150 switches/routers	
60.				

61.	Firewalls and Intrusion Detection Systems	Diagram-based explanation with configuration examples	Stallings (Network Security) Ch.9
62.	Virtual Private Networks (VPNs)	Concept mapping and protocol flow analysis	Stallings (Network Security) Ch.10
63.	Malware Detection	Threat analysis using real-world malware reports	Stallings (Network Security) Ch.7
64.	Lab 12: Port-Security	Hands-on using ICX7150 switches/routers	
65.			
66.	Virus Analysis	Demonstration-based learning with malware lifecycle	Stallings (Network Security) Ch.7
67.	SSL/TLS	Protocol walkthrough using handshake sequence analysis	Stallings (Network Security) Ch.7
68.	Secure Communication	Problem-based learning on secure channel design	Stallings (Network Security) Ch.1 & 7
69.	Practice Lab		
70.			
71.	IPsec and Internet Key Exchange (IKE)	Step-by-step protocol simulation	Stallings (Network Security) Ch.10
72.	DDoS Attacks	Scenario-based learning with attack-impact analysis	Stallings (Network Security) Ch.9
73.	Mitigation Strategies	Comparative study of defense mechanisms	Stallings (Network Security) Ch.9
74.	Mini Project Execution in Lab		
75.			

Assessment and Evaluation

Continuous Internal Evaluation (CIE): 70%	
Components of CIE	Weightage of each component
CIE-1	
- Quiz Graded Component 1	15 marks out of 70
- MOOC course Graded Component 2	5 marks out of 70
CIE-2	
- Theory	25 marks out of 70
CIE-3	
- Lab execution, verification of output & viva-Component 1	10 marks out of 70
- Final Project Submission- Component 2	15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of Exam	30%

Course Outcomes- Programme Outcomes Matrix

(3 = highly related, 2 = moderately related, 1 = slightly related, - = not related)

Sl. No	Course outcome	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
1.	Analyze the functionality of core transport and routing protocols, including TCP, RIP, and OSPF, to understand their roles in ensuring reliable and efficient network communication	3	3	2	2	-	-	2	-	-	-	2	-

2.	Explain multimedia transport protocols and assess the significance of Quality of Service (QoS) in supporting diverse network applications.	3	2	2	3	-	-	2	2	-	-	2	-
3.	Understand and apply cryptographic algorithms to enable secure communication by ensuring data confidentiality, integrity, and authentication.	3	3	-	3	-	-	2	2	-	2	2	-
4.	Implement IP and Web security mechanisms to design secure network environments and address real-world cybersecurity challenges.	2	3	2	2	2	2	3	2	3	2	2	-

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Component 1	Quiz will be conducted for 15 marks. Each right answer carries one mark. No negative marking for wrong answers.
	Component-2	Certificate of completion for 5 marks. The specified courses on the IBM SkillsBuild platform are to be uploaded to the designated Google Classroom. <i>(Refer Table-1)</i>

Table-1:

Graded Component-2 (5 Marks):

Assess students' ability to complete assigned courses on the IBM SkillsBuild platform and reflect on their learning.

Criteria	Excellent (5 Marks)	Satisfactory (3 Marks)	Needs Improvement (0 Marks)
Completion of Courses (5 Marks)	Successfully completes all specified courses on the MOOC platform before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by the team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Component 1	Continuous Lab assessment + Viva = 10 marks <i>(Refer Table-2)</i>
	Component 2	Course Project + Report = 15 marks <i>(Refer Table-3)</i>

Table-2

Lab Write-up and Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0 Mark)
1.	Understanding of the problem statement. (2 Marks)	Observations	Student exhibits a thorough understanding of the requirements and applies a suitable method to the problem. (2 M)	The student has a sufficient understanding of the requirements and applies a suitable method to the problem. (<2 M and >=1 M)	The student does not have a clear understanding of the requirements and is unable to apply a suitable method for the problem. (0 M)
2.	Execution (2 Marks)	Observations	The student demonstrates the execution of the network setup and shows performance efficiency. (2 M)	The student demonstrates the execution of the network setup with no performance metrics. (1 M)	The student has not demonstrated the network setup. (0 M)
3.	Results and Documentation on GCR (2 Marks)	Observations	Documentation with appropriate comments and all output screenshots is shown. (2 M)	Documentation with only few comments and only few output screenshots is shown. (1 M)	Documentation with no comments and no output screenshots is shown. (0 M)
Viva Voce Rubrics (Max: 4 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0 Mark)

1.	Conceptual Understanding (2 Marks)	Viva Voce	Explains thoroughly the network design and the components used along with related concepts. (2 M)	Adequately explains the network design and the components used with related concepts (1 M)	Unable to explain the network design and the components used (0 M)
2.	Communication of networking Concepts (2 Marks)	Viva Voce	Communicates the concept used in problem solving well. (2 M)	Sufficiently communicates the concepts used in problem solving. (1 M)	Unable to communicate the concepts used in problem. (0 M)

Table-4

Grading Rubrics for NS Group Projects (Total: 15 Marks)					
Sl. No	Criteria	Excellent	Very Good	Good	Needs Improvement
1.	Report Submission (3 Marks)	Comprehensive, well-organized, and insightful report that covers all aspects in depth. (3 M)	Thorough report with minor areas for improvement. (2 M)	Report covers all essentials but lacks detail or organization. (2 M)	Incomplete or significantly flawed report. (0-1 M)
2.	Project Demonstration (7 Marks)	Flawless execution demonstrating Networking problems (Hardware or software) techniques. (7 M)	Effective implementation with minor technical issues. (6M-4M)	Functional implementation with some errors or omissions. (3M-1M)	Lack of demonstration or significant errors present. (0 M)
3.	Presentation and Viva (5 Marks)	Outstanding delivery and depth of knowledge displayed in Q&A. (5 M)	Solid presentation and good comprehension shown in Q&A. (4 M)	Adequate presentation with some misunderstandings in Q&A. (2-3 M)	Presentation is incomplete or fails to address Q&A adequately. (0-1 M)